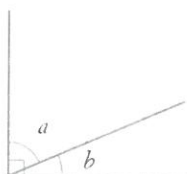




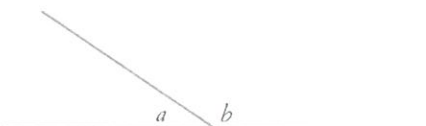
Circle the correct answer. Formula sheet and calculators may be used for this part.

- 1 Find the size of angle b if $a = 67^\circ$.



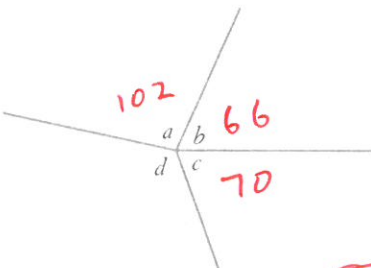
- A 23° ✓✓
B 45°
C 67°
D 77°

- 2 Find the size of angle b if $a = 34^\circ$.



- A 34°
B 124° ✓✓
C 56°
D 146°

- 3 Find the size of angle d if $a = 102^\circ$, $b = 66^\circ$ and $c = 70^\circ$.



- A 102°
B 122° ✓✓
C 117°
D 135°

- 4 What can be said about supplementary angles?

- A They are equal.
B They add up to 90° .
C They add up to 180° . ✓✓
D They add up to 360° .

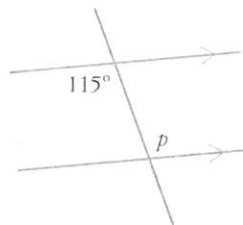
- 5 What can be said about vertically opposite angles?

- A They are equal. ✓✓
B They add up to 90° .
C They add up to 180° .
D They add up to 360° .

- 6 What angle is complementary to 30° ? ✓✓

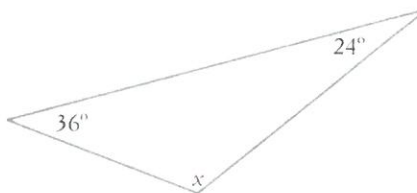
- A 30°
B 60°
C 90°
D 150°

- 7 Which statement is true about this diagram?



- A $p = 65$ (alternate angles)
B $p = 65$ (co-interior angles)
C $p = 115$ (alternate angles) ✓✓
D $p = 115$ (co-interior angles)

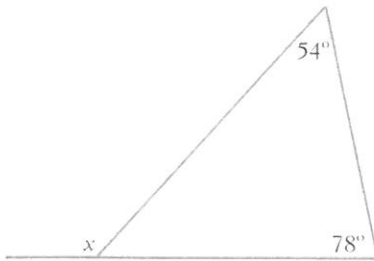
- 8 Find x .



- A $x = 36^\circ$
B $x = 60^\circ$
C $x = 120^\circ$ ✓✓
D $x = 150^\circ$

16

9 Find x .

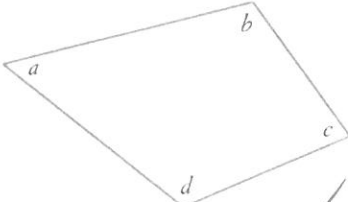


- A $x = 48^\circ$ B $x = 62^\circ$
 C $x = 102^\circ$ D $x = 132^\circ$

10 An isosceles triangle has an angle size of 70° . Its other two angles could be:

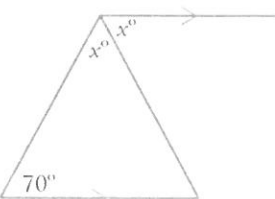
- A $55^\circ, 55^\circ$ B $70^\circ, 50^\circ$
 C $40^\circ, 40^\circ$ D $50^\circ, 60^\circ$

11 Which equation best describes the relationship between the four angles a , b , c and d ?



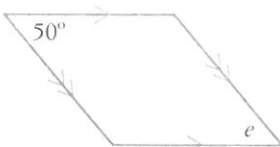
- A $a + b + c + d = 360^\circ$
 B $a + b + c + d = 180^\circ$
 C $a + c = b + d$
 D $d = a + b + c$

12 Find x .



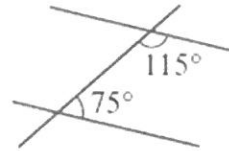
- A $x = 35^\circ$ B $x = 55^\circ$
 C $x = 70^\circ$ D $x = 110^\circ$

13 Find e .



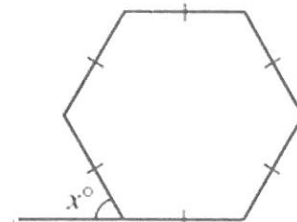
- A 40° B 50°
 C 60° D 130°

14 The pair of lines in the following diagram are:



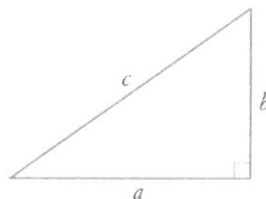
- A parallel, since cointerior angles are supplementary
 B parallel, since alternate angles are supplementary
 C parallel, since corresponding angles are supplementary
 D not parallel, since cointerior angles are not supplementary
 E not parallel, since alternate angles are not supplementary

15 The value of x in the following diagram is:



- A 30° B 36° C 45°
 D 60° E 120°

16 Which of the following sides is the hypotenuse of this triangle?



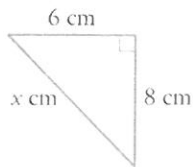
- A a B b
 C c D a or b

17 Which angle is opposite the hypotenuse in a right-angled triangle?

- A The complementary angle
 B The acute angle
 C The smallest angle
 D The right angle

18

18 Find x



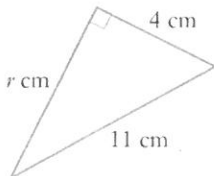
A $x = \sqrt{28}$

C $x = 12$

B $x = 10$

D $x = 14$

19 Find the value of r ?



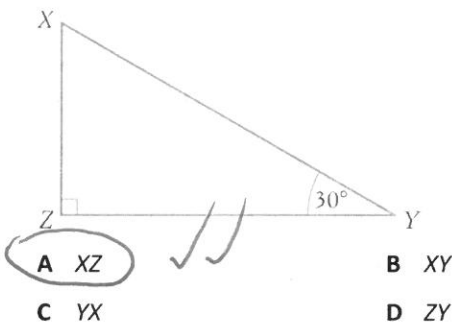
A $r \approx 11.7$

C $r = 8$

B $r \approx 10.25$

D $r \approx 6.25$

20 Which side of the triangle is the opposite?



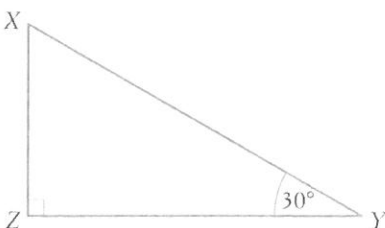
A XZ

C YX

B XY

D ZY

21 Which side of the triangle is the adjacent?



A XZ

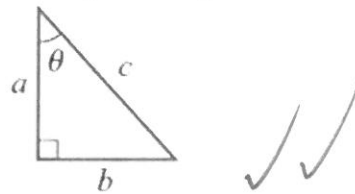
C YX

B XY

D ZY

Questions 22 and 23 refer to the following diagram:

22 For the given triangle:



A $\cos \theta = \frac{a}{b}$

B $\cos \theta = \frac{a}{c}$

C $\cos \theta = \frac{b}{c}$

D $\cos \theta = \frac{b}{a}$

E $\cos \theta = \frac{c}{a}$

23 For the given triangle:

A $\theta = \sin^{-1}\left(\frac{b}{c}\right)$

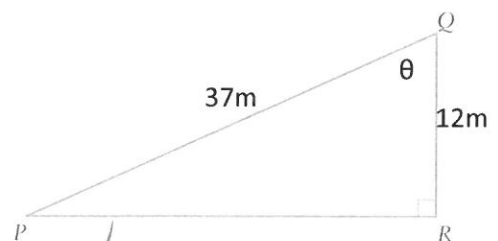
B $\theta = \sin^{-1}\left(\frac{b}{a}\right)$

C $\theta = \sin^{-1}\left(\frac{a}{c}\right)$

D $\theta = \sin^{-1}\left(\frac{a}{b}\right)$

E $\theta = \sin^{-1}\left(\frac{c}{b}\right)$

24 What is the size of $\angle Q$ if $QR = 12$ m and $PQ = 37$ m? Round your answer to the nearest degree.



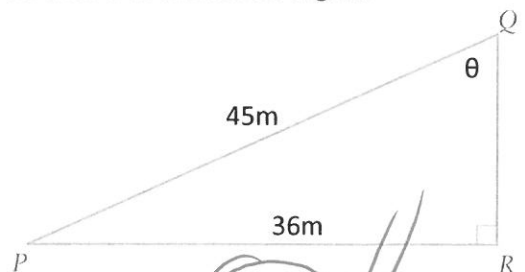
A 72°

B 71°

C 70°

D 19°

25 What is the size of $\angle Q$ if $PR = 36$ m and $PQ = 45$ m? Round your answer to the nearest degree.



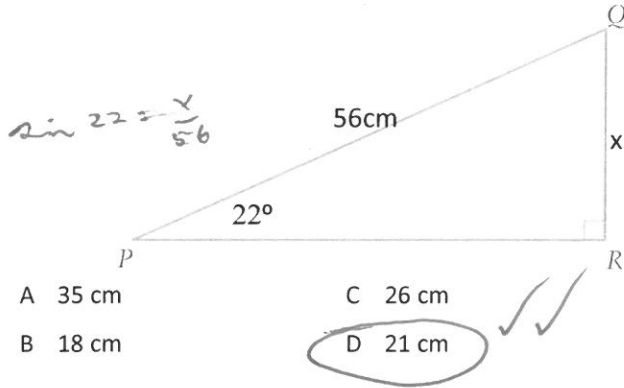
A 37°

B 36°

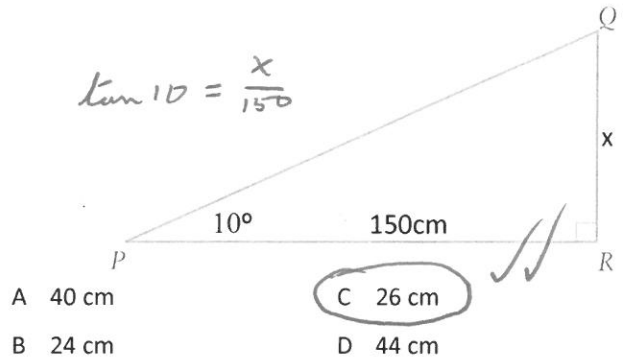
C 53°

D 54°

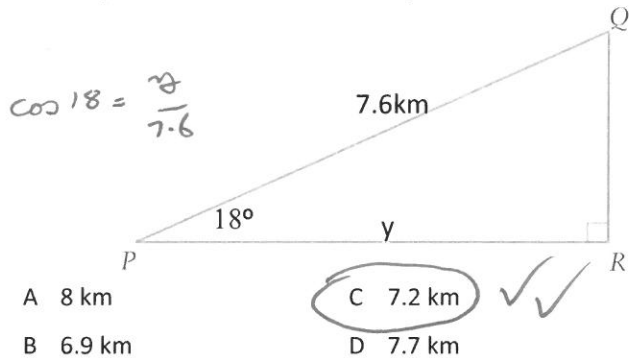
26 What is the length of x if $\angle P = 22^\circ$ and $PQ = 56$ cm?
Round your answer to the nearest whole number.



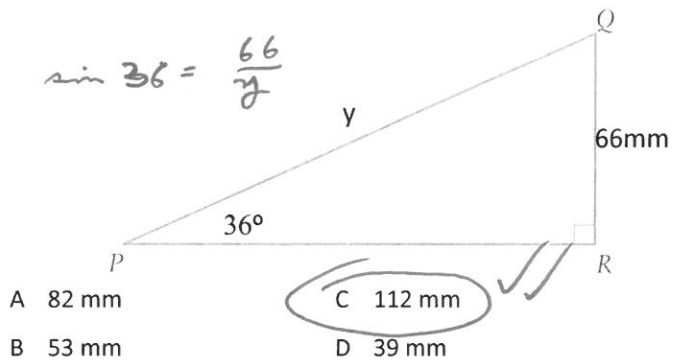
28 What is the length of x if $\angle P = 10^\circ$ and $PR = 150$ cm?
Round your answer to the nearest whole number.



27 What is the length of y if $\angle P = 18^\circ$ and $PQ = 7.6$ km?
Round your answer to one decimal place.



29 What is the length of y if $\angle P = 36^\circ$ and $QR = 66$ mm?
Round your answer to the nearest whole number.



30 What is the length of x if $\angle P = 16^\circ$ and $QR = 49$ km?
Round your answer to the nearest whole number.

